

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 2

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(210 - 12) / 210 \times 100$ (1) • = 94.3% (accept 94%) (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Conserves mineral reserves / extends their lifetime (1) • Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) (1) • Reduces mineral processing pollution (eg red mud from bauxite refining) (1) • Reduces waste disposal impacts / landfill volume (1) • Less CO₂ / climate change (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Identification of materials: slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) (1) • Separation of mixed materials: alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed; mixed wastes reduce quality (1) • Reduction in quality / contamination (eg paper mixed with food waste) (1) • Transport / collection: small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) (1) • Lack of consumer cooperation in separating wastes; waste losses to litter and general waste; labour costs (1)	4
01.4	<i>1 mark.</i> • Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	
02.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
	Model answer - Level 3 (8 marks)	
	Lasky's principle means that as grade falls, the quantity of metal available rises enormously, so depletion of high-grade deposits does not mean running out - it means the cut-off ore grade must fall, which depends on price, technology and energy costs. Better exploration finds what remains: high-resolution multispectral and infrared satellite sensors, gravimetry, magnetometry, seismic and resistivity surveys and portable field analysers narrow the search so expensive trial drilling is targeted. Mechanisation allows deeper underground mines and larger open-cast pits, and previously inaccessible deposits - polymetallic nodules and deep-sea sediments, and Antarctica if its protection were ever lifted - become possible, with serious environmental and legal problems.	

Question	Marking guidance	Marks
	The most sustainable route to low-grade ores is low-energy extraction: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement of copper, bacterial adsorption of rare earths and polymer adsorption from seawater all avoid the energy of crushing and smelting vast tonnages. Equally important is extending the use of metal already mined: recycling, Cradle-to-Cradle design so products can be disassembled, longer product lifetimes and substitution of scarce metals - the circular economy approach.	
	Each strategy has costs: deep-sea mining and Antarctica risk unique habitats, deeper mines need more energy, recycling is limited by dispersal and mixed alloys. The strategies that mimic natural low-temperature processes and keep metals in use are the most sustainable, and exploration and new mining should be the last resort rather than the first.	
	Weaker answer - Level 1 (3 marks)	
	When high grade ores run out we can mine lower grade ores instead, but this uses more energy. Better technology like satellites can find new deposits. We could mine the deep sea or Antarctica. Bacteria can be used to extract metals from low grade ore. Recycling metals means we do not need to mine as much. These strategies will help secure supplies of metals in the future.	
	What makes the difference	
	• Level 1 names most strategy types but Level 3 explains each (what the surveys measure, why low-temperature extraction matters, what Lasky's principle implies).	
	• Level 3 weighs environmental and energy costs and ranks the strategies; Level 1 offers no evaluation.	
	• Framing the answer with the concept of cut-off ore grade shows understanding rather than recall.	
02.2	<i>1 mark.</i> • All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	